

Monte Carlo Simulation for ISP Bandwidth Analysis

Modeling Internet Performance Under Congestion

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Outline

- Introduction
- Literature Review
- Objectives
- Methods / Algorithm
- Results & Discussion
- Conclusion
- Acknowledgement
- Q&A

Introduction

- ISP speeds vary from advertised values.
- Peak-hour congestion affects performance.
- Real ISP data is difficult to access.
- Simulation-based modeling is required.

Literature Review

- Queueing theory explains congestion
- Simulation for system performance analysis
- Monte Carlo models stochastic systems
- Distributed networks manage dynamic traffic

Objectives

- Model ISP bandwidth using synthetic data.
- Simulate using Monte Carlo methods.
- Compare mean and variance.
- Analyze hourly patterns and convergence.

Methods Overview

- Generate multi-user dataset (24 hrs)
- Run Monte Carlo simulation
- Model user load and capacity
- Compare real vs simulated

Algorithm

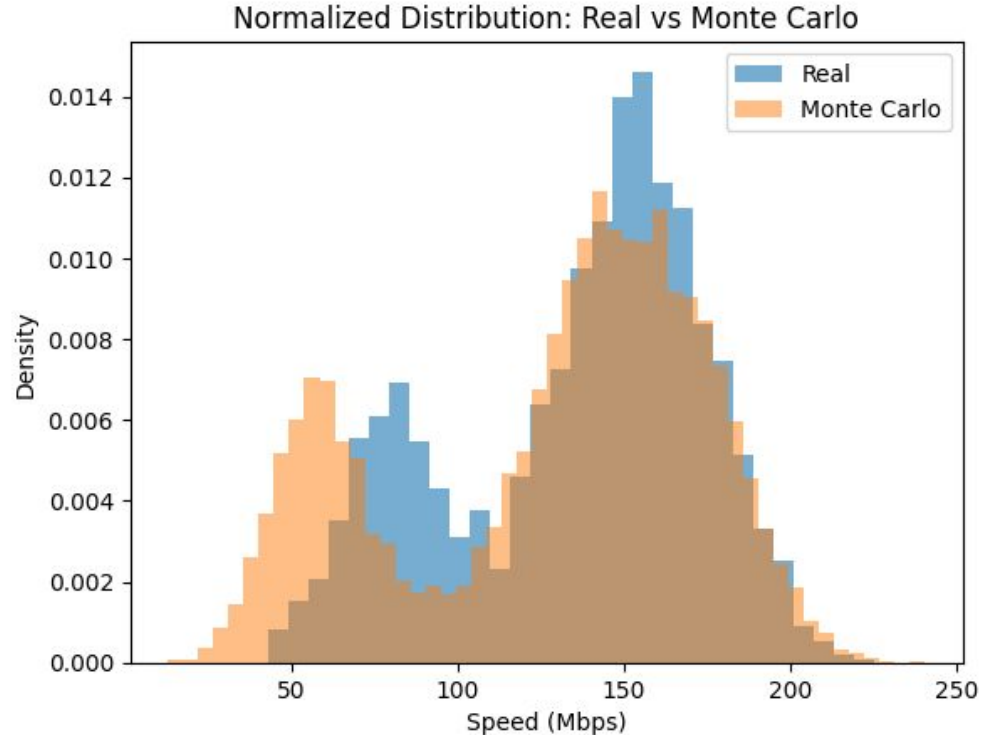
- Select random hour
- Generate users (Poisson)
- Generate capacity (Normal)
- Compute bandwidth
- Repeat for many iterations

Results Overview

- Distribution comparison
- Hourly mean comparison
- Hourly variance comparison
- Convergence analysis
- Sampling stability

Distribution Comparison

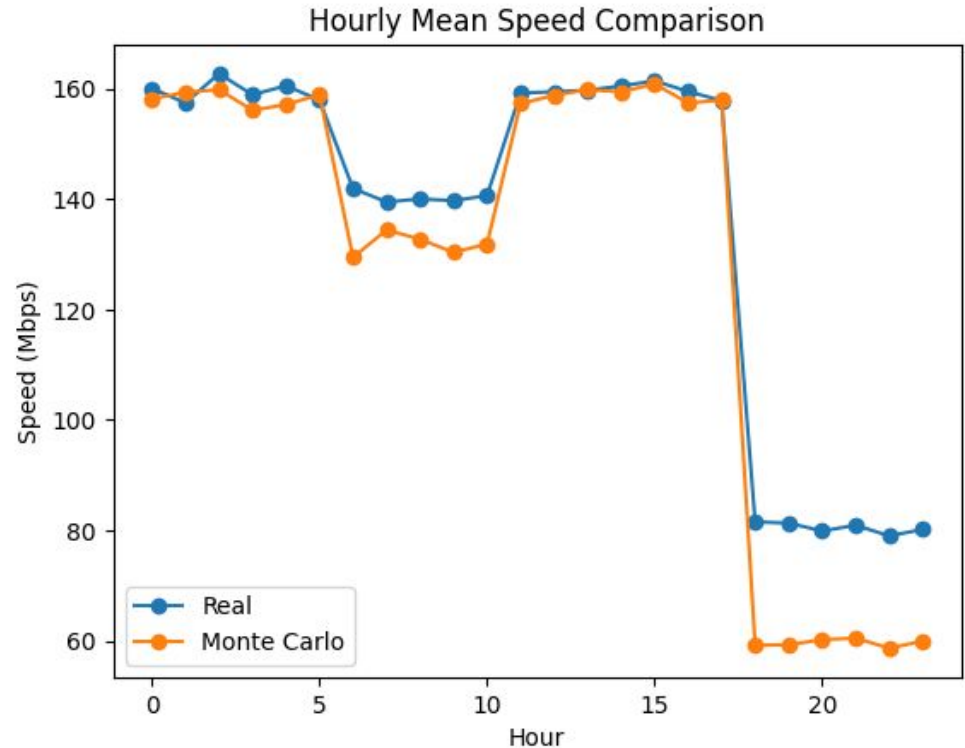
- Normalized density comparison
- Captures peak and off-peak behavior
- Both datasets exhibit similar shapes, confirming that the simulation captures realistic bandwidth patterns, including peak-hour degradation and off-peak stability.



Hourly Mean Comparison

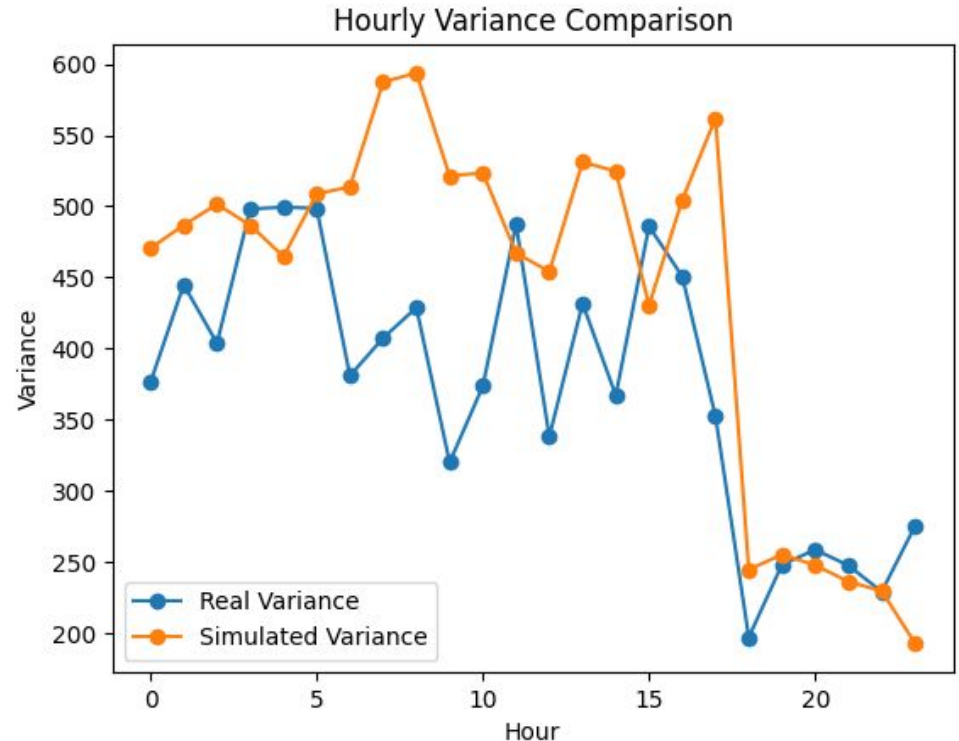
- Higher speeds during off-peak hours.
- Moderate speeds in the morning
- Significant drop during peak evening hours

The simulation closely follows this trend, demonstrating its ability to model time-dependent congestion.



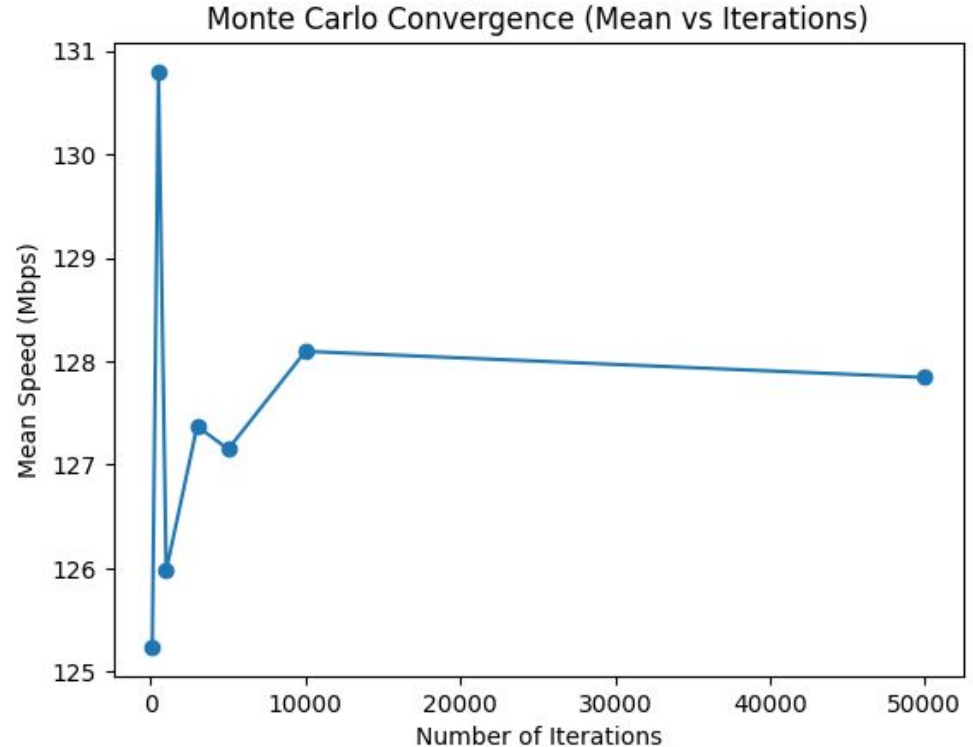
Hourly Variance

- Variance increases during peak hours
- Indicates instability
- Simulation replicates variability



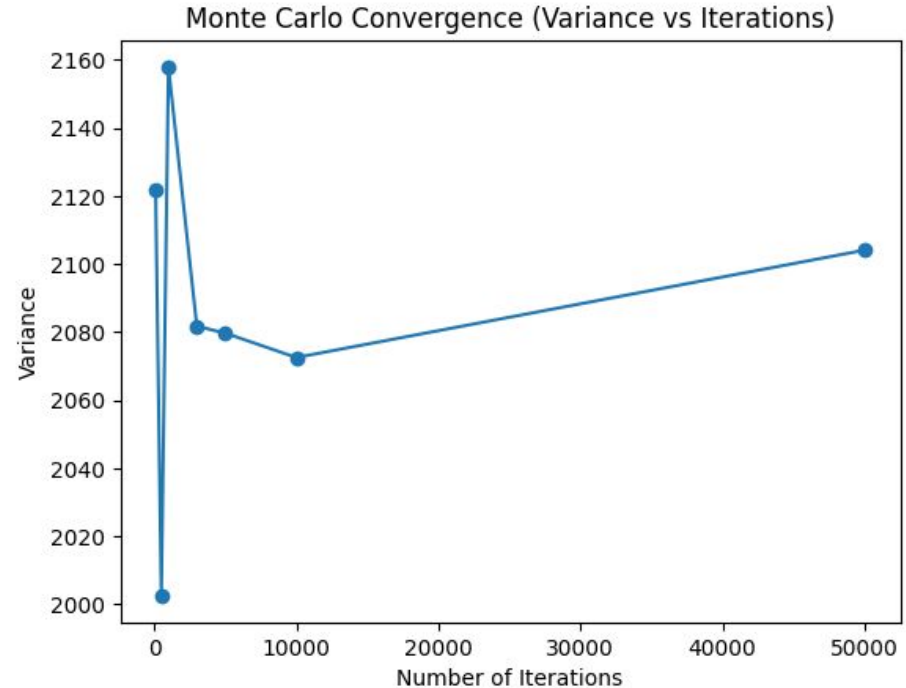
Mean Convergence

- Monte Carlo stabilizes with iterations
- Higher samples = better estimate



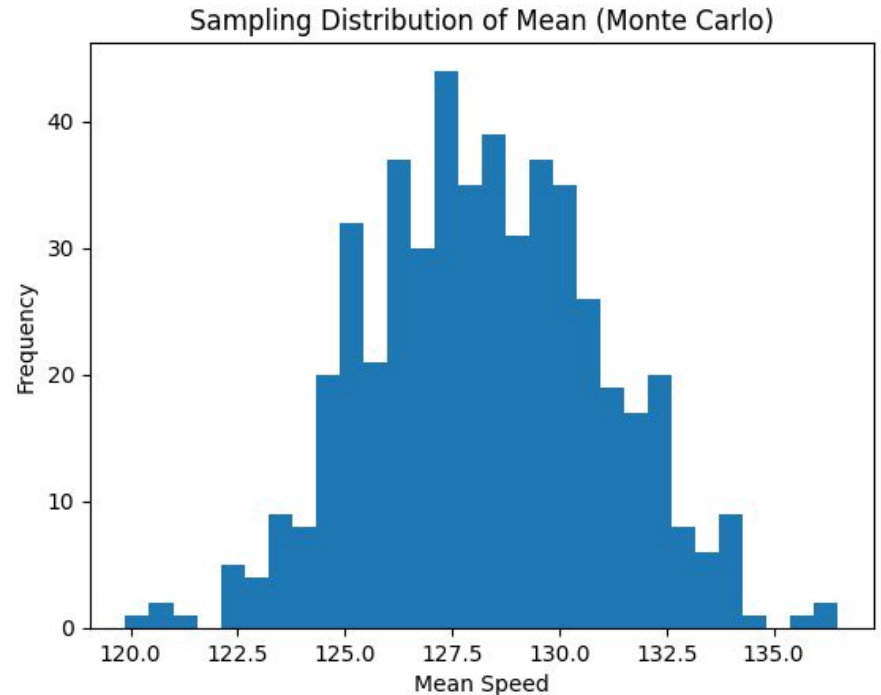
Variance Convergence

- Variance stabilizes with more iterations
- Confirms simulation reliability



Sampling Stability

- Approximately normal and centered around the overall mean.
- Indicates stability and consistency of the simulation results.



Key Findings

- Simulation matches realistic patterns
- Peak-hour congestion captured
- Variability modeled effectively
- Monte Carlo converges reliably

Conclusion

- Monte Carlo effective for ISP modeling
- Captures congestion and variability

Limitations

- Synthetic dataset
- Assumptions on user behavior
- No latency/packet loss

Future Work

- Use real ISP data
- Add latency and packet loss
- Multi-ISP comparison

References

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Thank You

Any queries?